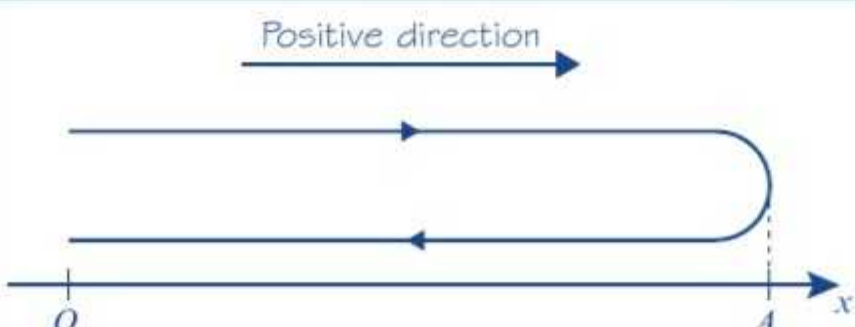




a $a = -2.5$, $u = 15$, $s = 0$, $t = ?$

$$s = ut + \frac{1}{2}at^2$$

$$0 = 15t + \frac{1}{2} \times (-2.5) \times t^2$$

$$0 = 60t - 5t^2$$

$$= 5t(12 - t)$$

$$t = 0, t = 12$$

The particle P returns to O after 12 s.

b $a = -2.5$, $u = 15$, $v = 0$, $s = ?$

$$v^2 = u^2 + 2as$$

$$0^2 = 15^2 + 2 \times (-2.5) \times s$$

$$5s = 15^2 = 225$$

$$s = \frac{225}{5} = 45$$

The distance $OA = 45$ m.

The total distance travelled by P is $2 \times 45 \text{ m} = 90 \text{ m}$.

Problem-solving

Before you start, draw a sketch so you can see what is happening. The particle moves through O with a positive velocity. As it is decelerating it slows down and will eventually have zero velocity at a point A , which you don't yet know. As the particle is still decelerating, its velocity becomes negative, so the particle changes direction and returns to O .

When the particle returns to O , its displacement (distance) from O is zero.

Multiply by 4 to get whole-number coefficients.

At the furthest point from O , labelled A in the diagram, the particle changes direction. At that point, for an instant, the particle has zero velocity.

In the 12 s the particle has been moving it has travelled to A and back. The total distance travelled is twice the distance OA .

Exercise 2E

SKILLS PROBLEM-SOLVING

- 1 A particle is moving in a straight line with constant acceleration 2.5 m s^{-2} . It passes a point A with velocity 3 m s^{-1} and later passes through a point B , where $AB = 8 \text{ m}$. Find the velocity of the particle as it passes through B .
- 2 A car is accelerating at a constant rate along a straight horizontal road. Travelling at 8 m s^{-1} , it passes a lamp post and 6 s later it passes a sign. The distance between the lamp post and the sign is 60 m. Find the acceleration of the car.
- 3 A cyclist travelling at 12 m s^{-1} applies her brakes and comes to rest after travelling 36 m in a straight line. Assuming that the brakes cause the cyclist to decelerate uniformly, find the deceleration.
- 4 A train is moving along a straight horizontal track with constant acceleration. The train passes a signal with a velocity of 54 km h^{-1} and a second signal with a velocity of 72 km h^{-1} . The distance between the two signals is 500 m. Find, in m s^{-2} , the acceleration of the train.
- 5 A particle moves along a straight line, with constant acceleration, from a point A to a point B where $AB = 48 \text{ m}$. At A the particle has velocity 4 m s^{-1} and at B it has velocity 16 m s^{-1} . Find:
 - a the acceleration of the particle
 - b the time the particle takes to move from A to B .

- E/P** 14 Two particles P and Q are moving along the same straight horizontal line with constant accelerations 2 m s^{-2} and 3.6 m s^{-2} respectively. At time $t = 0$, P passes through a point A with speed 4 m s^{-1} . One second later Q passes through A with speed 3 m s^{-1} , moving in the same direction as P .
- Write down expressions for the displacements of P and Q from A , in terms of t , where t seconds is the time after P has passed through A . (2 marks)
 - Find the value of t where the particles meet. (3 marks)
 - Find the distance of A from the point where the particles meet. (3 marks)
- Problem-solving**
When P and Q meet, their displacements from A are equal.
- E/P** 15 In an orienteering competition, a competitor moves in a straight line past three checkpoints, P , Q and R , where $PQ = 2.4 \text{ km}$ and $QR = 11.5 \text{ km}$. The competitor is modelled as a particle moving with constant acceleration. She takes 1 hour to travel from P to Q and 1.5 hours to travel from Q to R . Find:
- the acceleration of the competitor
 - her speed at the instant she passes P . (7 marks)

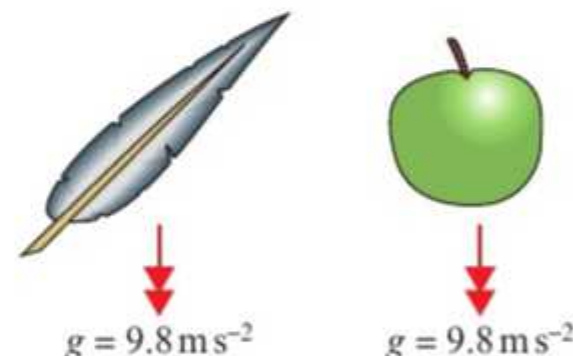
2.6 Vertical motion under gravity

You can use the formulae for constant acceleration to model an object moving vertically under gravity.

- The force of gravity causes all objects to accelerate toward the Earth. If you ignore the effects of air resistance, this acceleration is constant. It does not depend on the mass of the object.

As the force of gravity does not depend on mass, this means that in a vacuum an apple and a feather would both accelerate downward at the same rate.

On Earth, the acceleration due to gravity is represented by the letter g and is approximately 9.8 m s^{-2} .



The actual value of the acceleration can vary by very small amounts in different places due to the changing radius of the earth and height above sea level.

- An object moving vertically under gravity can be modelled as a particle with a constant downward acceleration of $g = 9.8 \text{ m s}^{-2}$.

When solving problems about vertical motion you can choose the positive direction to be either upward or downward. Acceleration due to gravity is always downward, so if the positive direction is upward then $g = -9.8 \text{ m s}^{-2}$.

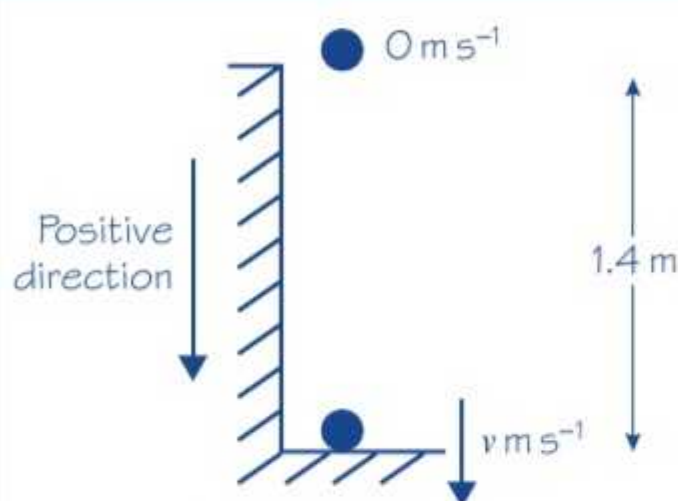
Watch out In mechanics questions you will always use $g = 9.8 \text{ m s}^{-2}$ unless a question specifies otherwise. However, if a different value of g is specified (e.g. $g = 10 \text{ m s}^{-2}$ or $g = 9.81 \text{ m s}^{-2}$) the degree of accuracy in your answer should be chosen to be consistent with this value.

Notation The total time that an object is in motion from the time it is projected (thrown) upward to the time it hits the ground is called the **time of flight**. The initial speed is sometimes called the **speed of projection**.

Example 11**SKILLS** PROBLEM-SOLVING

A book falls off the top shelf of a bookcase. The shelf is 1.4 m above a wooden floor. Find:

- the time the book takes to reach the floor
- the speed with which the book strikes the floor.



a

$$s = 1.4$$

$$a = +9.8$$

$$u = 0$$

$$t = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$1.4 = 0 + \frac{1}{2} \times 9.8 \times t^2$$

$$t^2 = \frac{1.4}{4.9} = 0.2857\dots$$

$$t = \sqrt{0.2857\dots} = 0.5345\dots$$

The time taken for the book to reach the floor is 0.53 s, to two significant figures.

b

$$s = 1.4$$

$$a = 9.8$$

$$u = 0$$

$$v = ?$$

$$v^2 = u^2 + 2as$$

$$= 0^2 + 2 \times 9.8 \times 1.4 = 27.44$$

$$v = \sqrt{27.44} = 5.238\dots \approx 5.2$$

The book hits the floor with speed 5.2 m s⁻¹, to two significant figures.

Model the book as a particle moving in a straight line with a constant acceleration of magnitude 9.8 m s⁻².

As the book is moving downward throughout its motion, it is sensible to take the downward direction as positive.

You have taken the downward direction as positive and gravity acts downward. Here the acceleration is positive.

Assume the book has an initial speed of zero.

Choose the formula without v .

Solve the equation for t^2 and use your calculator to find the positive square root.

Give the answer to two significant figures to be consistent with the degree of accuracy used for the value of g .

Choose the formula without t .

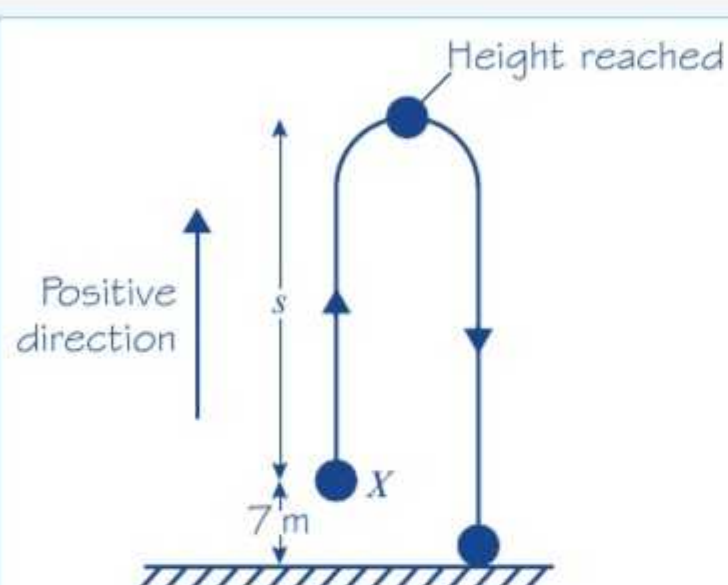
Use unrounded values in your calculations, but give your final answer correct to two significant figures.

Example 12

SKILLS PROBLEM-SOLVING

A ball is projected vertically upward, from a point X which is 7 m above the ground, with speed 21 m s^{-1} . Find:

- the greatest height above the ground reached by the ball
- the time of flight of the ball.



a

$$u = 21$$

$$v = 0$$

$$a = -9.8$$

$$s = ?$$

$$v^2 = u^2 + 2as$$

$$0^2 = 21^2 + 2 \times (-9.8) \times s = 441 - 19.6s$$

$$s = \frac{441}{19.6} = 22.5$$

$$(22.5 + 7) \text{ m} = 29.5 \text{ m}$$

Greatest height is 30 m (2 s.f.)

b

$$s = -7$$

$$u = 21$$

$$a = -9.8$$

$$t = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$-7 = 21t - 4.9t^2$$

$$4.9t^2 - 21t - 7 = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-21) \pm \sqrt{(-21)^2 - 4 \times 4.9 \times (-7)}}{2 \times 4.9}$$

$$= \frac{21 \pm \sqrt{578.2}}{9.8} \approx \frac{21 \pm 24.046}{9.8}$$

$$t \approx 4.5965,$$

$$\text{or } t \approx -0.3108$$

Time of flight is 4.6 s (2 s.f.)

Problem-solving

In this sketch the upward and downward motions have been sketched side by side. In reality they would be on top of one another, but drawing them separately makes it easier to see what is going on. Remember that X is 7 m above the ground, so mark this height on your sketch.

At its highest point, the ball is turning around. For an instant, it is neither going up nor down, so its velocity is zero.

22.5 m is the distance the ball has moved above X , but X is 7 m above the ground. You must add on another 7 m to get the greatest height above the ground reached by the ball.

The time of flight is the total time that the ball is in motion from the time that it is projected to the time that it stops moving. Here the ball will stop when it hits the ground. The point where the ball hits the ground is 7 m below the point from which it was projected so $s = -7$.

Online Use your calculator to check solutions to quadratic equations quickly.



Rearrange the equation and use the quadratic formula.

Take the positive answer and round to two significant figures.

Example

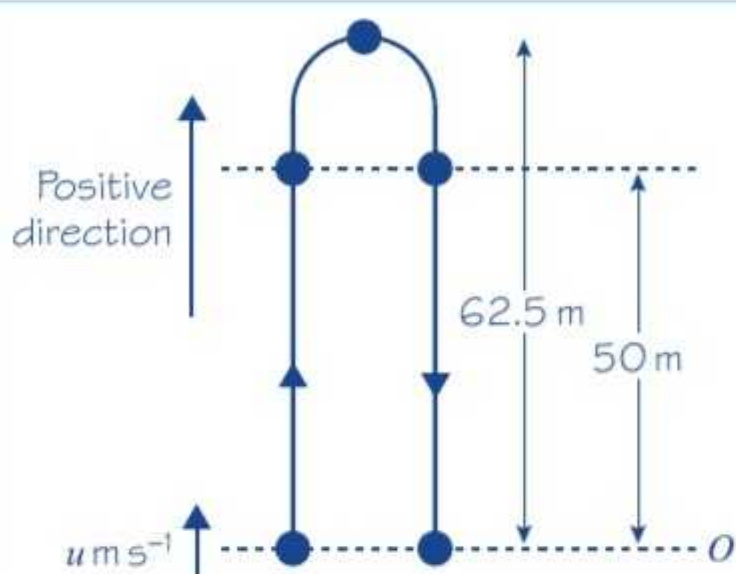
13

SKILLS

PROBLEM-SOLVING

A particle is projected vertically upward from a point O with speed $u \text{ m s}^{-1}$. The greatest height reached by the particle is 62.5 m above O . Find:

- the value of u
- the total time for which the particle is 50 m or more above O .



a $v = 0$

$$s = 62.5$$

$$a = -9.8$$

$$u = ?$$

$$v^2 = u^2 + 2as$$

$$0^2 = u^2 + 2 \times (-9.8) \times 62.5$$

$$u^2 = 1225$$

$$u = \sqrt{1225} = 35 \text{ m s}^{-1}$$

b $s = 50$

$$u = 35$$

$$a = -9.8$$

$$t = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$50 = 35t - 4.9t^2$$

$$4.9t^2 - 35t + 50 = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{35 \pm \sqrt{(35)^2 - 4 \times 4.9 \times 50}}{9.8}$$

$$= \frac{35 \pm \sqrt{245}}{9.8} \approx \frac{35 \pm 15.6525}{9.8}$$

$$t \approx 5.1686..., \text{ or } t \approx 1.9742...$$

$$(5.1686...) - (1.9742...) \approx 3.194$$

Particle is 50 m or more above O for 3.2 s
(2 s.f.)

The particle will pass through the point 50 m above O twice: once on the way up and once on the way down.

There is no t , so you choose the formula without t .

In this part, you obtain an exact answer, so there is no need for approximation.

Two values of t need to be found: one on the way up and one on the way down.

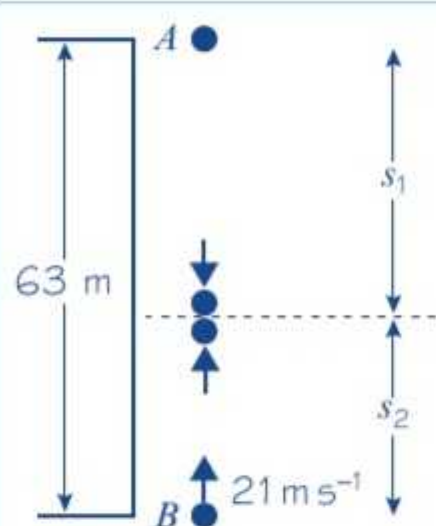
Write the equation in the form $ax^2 + bx + c = 0$ and use the quadratic formula.

Between these two times the particle is always more than 50 m above O . You find the total time for which the particle is 50 m or more above O by finding the difference of these two values.

Example 14

SKILLS PROBLEM-SOLVING

A ball A falls vertically from rest at the top of a tower 63 m high. At the same time as A begins to fall, another ball B is projected vertically upward from the bottom of the tower with speed 21 m s^{-1} . The balls collide. Find the distance of the point where the balls collide from the bottom of the tower.



For A , the motion is downward

$$u = 0$$

$$a = 9.8$$

$$s = ut + \frac{1}{2}at^2$$

$$s_1 = 4.9t^2$$

For B , the motion is upward

$$u = 21$$

$$a = -9.8$$

$$s = ut + \frac{1}{2}at^2$$

$$s_2 = 21t - 4.9t^2$$

The height of the tower is 63 m.

$$s_1 + s_2 = 63$$

$$4.9t^2 + (21t - 4.9t^2) = 63$$

$$21t = 63$$

$$t = 3$$

$$s_2 = 21t - 4.9t^2$$

$$= 21 \times 3 - 4.9 \times 3^2 = 18.9$$

The balls collide 19 m from the bottom of the tower, to two significant figures.

Problem-solving

You must take special care with problems where objects are moving in different directions. Here A is moving downward and you will take the acceleration due to gravity as positive. However, B is moving upward so for B the acceleration due to gravity is negative.

You cannot find s_1 at this stage. You have to express it in terms of t .

As B is moving upward, the acceleration due to gravity is negative.

You now have expressions for s_1 and s_2 in terms of t .

Adding together the two distances gives the height of the tower. You can write this as an equation in t .

You have found t but you were asked for the distance from the bottom of the tower. Substitute your value for t into your equation for s_2 .

Exercise 2F

SKILLS PROBLEM-SOLVING

- 1 A cliff diver jumps from a point 28 m above the surface of the water. Modelling the diver as a particle moving freely under gravity with initial velocity 0, find:
 - a the time taken for the diver to hit the water
 - b the speed of the diver when he hits the water.
- 2 A particle is projected vertically upward with speed 20 m s^{-1} from a point on the ground. Find the time of flight of the particle.